

DOCUMENT #6 OF 10 — PROJECT EXECUTION & ROADMAP

Implementation Plan & Engineering Roadmap

Complete Phased Delivery Blueprint: Project Setup, Auth, Database, UI/UX, AI Integrations & Deployment

Execution Model	Primary Stack	AI Toolchain	Document Version
Agile 7-Stage Sprint Delivery	React 19 + TypeScript + Vite + Supabase	Google Gemini / Claude API + Web Speech	v1.0 (Production Roadmap)

1. Implementation Roadmap Overview

The implementation plan is structured across seven consecutive, milestone-driven phases designed for zero regression, rapid development, and strict validation of business logic.

Phase	Milestone Name	Core Objectives & Engineering Deliverables	Estimated Timeline
Phase A	Project Structure & Init	Monorepo setup with Vite + React 19 + TypeScript, Git branch rules, ESLint, Prettier, environment variable schema (.env.example).	Sprint 1 (Days 1–2)
Phase B	Auth & Session Layer	Supabase Auth integration, OAuth providers (Google/Apple), session storage, Protected Route wrappers in React Router.	Sprint 1 (Days 3–4)
Phase C	Database & RLS Setup	PostgreSQL schema migration, 8 relational tables, Foreign Keys, B-tree indexes, Row-Level Security policies, seed demo data.	Sprint 1 (Days 5–6)
Phase D	Core UI/UX & Layout	Vanilla CSS Glassmorphism design system, persistent dark sidebar (#090C1F), topbar, 5 core screens wireframe layout, reusable UI kit.	Sprint 2 (Days 7–9)
Phase E	Core AI Features & Logic	Drag-and-drop resume ingestion, parsing pipeline, Google Gemini/Claude scoring engine, match score % matrix, ATS audit logic.	Sprint 2 (Days 10–12)
Phase F	Third-Party Integrations	Supabase Storage bucket upload with signed URLs, Web Speech voice-to-text integration, payment gateway webhook triggers.	Sprint 3 (Days 13–14)
Phase G	Testing & Production Deploy	End-to-end user journey tests, security scan, bundle optimization, Vercel/Netlify frontend edge deploy + Supabase Cloud production release.	Sprint 3 (Days 15–16)

2. Detailed Phase Breakdowns

Phase A: Project Structure & Initialization

Deliverables & Standards:

- Initialize React 19 TypeScript project using Vite template (`npm create vite@latest -- --template react-ts`).
- Establish repository branching strategy: `main` (production), `develop` (staging), and `feature/*` branches.
- Configure `.env.local` supporting `VITE_SUPABASE_URL`, `VITE_SUPABASE_ANON_KEY`, and `VITE_GEMINI_API_KEY`.

Phase B: Authentication, Authorization & Session Management

Deliverables & Standards:

- Create Supabase Client singleton (`supabase.ts`) using `@supabase/supabase-js`.
- Build unified `AuthProvider` React context handling user state, token refresh, and login/logout callbacks.
- Implement route guards (`<ProtectedRoute />`) restricting unauthenticated visitors from accessing dashboard views.

Phase C: Database Setup, Schema & Seed Initial Data

Deliverables & Standards:

- Execute DDL scripts creating 8 core tables (users, projects, tasks, candidates, comments, notifications, sessions, payments).
- Enable Row Level Security (RLS) policies on all tables ensuring multi-tenant project and resume isolation.
- Seed initial mockup records (e.g., sample candidate profiles: *Rahul Sharma*, *Priya Verma*, etc.) for local verification.

Phase D: Core UI/UX, Glassmorphism Styling & 5 Screens

Deliverables & Standards:

- Implement Vanilla CSS design tokens (`--glass-bg`, `--glass-border`, `backdrop-filter`) without external UI overhead.
- Build persistent sidebar navigation with active glowing purple pill indicators (`#4338ca`).
- Implement the 5 verified screen wireframes: 1. *Dashboard (with 4 KPI cards)*, 2. *Upload & Input*, 3. *Analysis Result*, 4. *Candidate Ranking*, and 5. *Resume Summary*.

Phase E: Core Features & Business Logic Implementation

Deliverables & Standards:

- Build multi-file drag-and-drop resume upload zone with client-side format & size validation (PDF/DOCX $\leq$ 5MB).
- Connect Supabase Edge Functions to Google Gemini / Claude APIs to extract entities (Education, Experience, Skills).
- Compute exact and semantic skill overlap %, ATS compatibility scoring, duplicate detection, and candidate rank ordering.

Phase F: Integrations (Storage, Voice AI & Notifications)

Deliverables & Standards:

- Configure Supabase Storage bucket (`resumes-vault`) with private signed URL access.
- Integrate native browser Web Speech API for hands-free conversational interview responses.
- Implement real-time notification alerts (Supabase Realtime channels) on completed batch analyses.

Phase G: Testing, Hardening & Production Deployment

Deliverables & Standards:

- Execute cross-browser validation, error boundary audits, and zero-state / offline edge case checks.
- Deploy React 19 SPA to **Vercel / Netlify** with automatic edge caching and gzip/brotli compression.
- Configure production database connection pooling on **Supabase Cloud** with daily automated snapshots.